

YASH MAHESH BANGERA

+91 9892279995 | yashmaheshbangera.work@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with 5+ years of experience architecting and deploying production-grade Machine Learning, Generative AI, and MLOps solutions on AWS for Fortune 500 enterprises across healthcare, finance, IT, and manufacturing. Proven record designing agentic AI systems, RAG pipelines, and LLM-powered automation using Amazon Bedrock, SageMaker, LangChain, and LangGraph — driving up to 80% efficiency gains and a 60% reduction in post-deployment issues. AWS-certified (Machine Learning Engineer, AI Practitioner, Solutions Architect) with deep expertise translating ambiguous enterprise requirements into scalable, secure, customer-facing AI solutions. Actively seeking AI Engineer, Machine Learning Engineer, Data Scientist, and Forward Deployed Engineer opportunities in the UAE and Singapore.

EDUCATION

M.S., Data Science — Rochester Institute of Technology

Rochester, NY | May 2022

B.Tech, Computer Science & Engineering — Vellore Institute of Technology

Chennai, India | May 2019

TECHNICAL SKILLS

- Generative AI & LLM Engineering:** Amazon Bedrock, Bedrock AgentCore, LangChain, LangGraph, Strands Agents, LlamaIndex, HuggingFace Transformers, RAG Architectures, vLLM, LangSmith, RAGAS, Vector DBs (FAISS, Pinecone, pgvector, ChromaDB, Qdrant)
- Machine Learning & Deep Learning:** Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, Keras, Computer Vision (OpenCV), NLP (NLTK, SpaCy, Gensim)
- MLOps & Cloud Infrastructure:** Amazon SageMaker, SageMaker Distributed Training (SMP v2), MLflow, Airflow, Kubeflow, Kafka, Docker, Kubernetes, AWS CDK, Terraform, GitHub Actions, GitLab CI/CD
- AWS Services:** SageMaker, SageMaker Model Cards, SageMaker Geospatial ML, SageMaker Ground Truth, Bedrock, Textract, Transcribe Medical, Rekognition, Comprehend, Personalize, A2I, API Gateway, Lambda, S3, Glue, Athena, DynamoDB, CloudFront, IAM, KMS, VPC, SNS, SQS, CloudWatch
- Programming & Data:** Python, R, Java, C/C++, MATLAB, SQL, PostgreSQL, PySpark, Pandas, NumPy, Jupyter, Tableau
- Full-Stack & Dev Tools:** React.js, TypeScript, FastAPI, Flask, Streamlit, Git, VS Code, PyCharm, Kiro IDE

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- AWS Certified AI Practitioner
- AWS Certified Solutions Architect – Associate
- Machine Learning – Stanford University (Coursera)

PROFESSIONAL EXPERIENCE

Associate Delivery Consultant, AI/ML — Amazon Web Services (AWS)

Dec 2025 – June 2026

- Partner directly with enterprise customers' product, engineering, and data science teams to translate ambiguous requirements into production-grade AI/ML solutions using SageMaker and Bedrock, unlocking 20–30% potential revenue optimization — operating as a forward-deployed technical advisor embedded in customer environments.
- Architected an intelligent document processing pipeline (Amazon Bedrock, Lambda, S3, SQS, OpenSearch, OCR, NLP) enabling workforce re-allocation, cutting costs up to 40%, and delivering 70–80% efficiency gains.
- Unified classical ML, GenAI, and LLMops practices with AWS AI/ML services; automated deployment via AWS CDK/Terraform and CI/CD, reducing deployment time 50% while strengthening security posture.
- Accelerated PoC delivery using AI-powered code generation (Kiro IDE), cutting project development time from 4 weeks to 1.5 weeks.

Cloud Support Engineer I, SageMaker, Vision & Other Services — Amazon Web Services (AWS)

Jun 2022 – Dec 2025

- Owned resolution of 500+ critical, escalated SageMaker issues for Fortune 500 customers across healthcare, finance, IT, and manufacturing — delivering real-time inference, latency optimization, and custom model hosting solutions that cut post-deployment issues 60%.
- Ranked #2 globally within Amazon SageMaker Support Operations for case complexity and resolution speed; sustained 95%+ CSAT across 7+ consecutive quarters and co-authored standard operating procedures adopted team-wide.
- Ran A/B tests on SageMaker endpoints using an 80/20 production-shadow traffic split, identifying a 100ms latency increase and 20% error rate rise in the new variant that drove a decision to hold back the rollout.
- Ran distributed training jobs on Amazon SageMaker (SMP v2, custom Docker container) using tensor parallelism to split model layers across multiple GPUs, tuning hyperparameters to cut overall job execution time 50%.

Teaching & Research Assistant — Rochester Institute of Technology

Mar 2021 – Dec 2021

- Mentored 30+ students in Data Science Fundamentals and Python, driving 90% proficiency in building and deploying ML models (K-NN, Neural Networks, Decision Trees) through targeted feedback achieving 95%+ accuracy.
- Engineered NLP feature pipelines (Pandas, NLTK) across 500K+ data points for two linguistic research studies, producing 10 new features (word embeddings, GloVe, TF-IDF) and cutting model training time 35% while lifting accuracy to 85%+.

Lead Machine Learning Engineer — Omdena, Inc.

Oct 2019 – Jul 2020

- Led a computer vision team using ResNet to detect Amazon Rainforest wildfires, cutting false positives 10–33% via Gaussian blur removal and reaching 95–97% model accuracy.
- Engineered 10 statistical features and built a DBSCAN clustering model (scikit-learn) powering an ethical credit-scoring system for underbanked Indonesian borrowers.
- Led a team of 28 engineers to analyze COVID-19 policy impacts on vulnerable populations (Tableau, SQL, NLTK); findings presented at the UN AI for Good Global Summit and showcased by NASDAQ in Times Square.

Research & Development Intern — AI Square Global Solutions LLP

May 2019 – Jul 2020

- Built a healthcare-sector NLP dataset from 10,000+ scraped news articles and applied 7 HuggingFace summarization models (evaluated via ROUGE/BLEU), lifting downstream analysis accuracy 15% and customer satisfaction 50%.

KEY PROJECTS

Self-Hosted RAG Pipeline for Financial Document Analysis — Qdrant, vLLM, BGE Models, Phi-4-mini-instruct

Aug – Sep 2026

- Built a fully private, self-hosted RAG pipeline serving open-weight embedding (BAAI/bge-large-en-v1.5), reranker (BAAI/bge-reranker-base), and generator (Microsoft Phi-4-mini-instruct) models via vLLM on a GPU-accelerated WSL2 environment, with no dependency on public cloud AI APIs.
- Applied domain-aware parsing and structure-aware hybrid chunking to 10-K and 10-Q SEC EDGAR filings across 10 tickers, producing ~87,000 chunks embedded into a self-hosted Qdrant vector database with rich metadata.
- Evaluated pipeline quality using RAGAS metrics and Hit Rate@K, achieving ~84.5% faithfulness and recall for the financial-domain use case.

Intelligent Document Processing Pipeline — AWS CDK, Textract, Comprehend, Bedrock

Jan – Mar 2026

- Cut infrastructure provisioning time 60% and operational costs 32% by architecting a fully automated CI/CD pipeline (GitLab, SQS, SNS) that eliminated manual deployments end-to-end.
- Engineered a serverless OCR/NLP pipeline (Python, AWS CDK, Textract, Comprehend, Bedrock) extracting structured insights — summaries, entities, keywords — from thousands of documents across 10 formats.
- Achieved 100% fault-tolerant processing by parallelizing the workflow with SQS and Lambda, including automatic retry and partial batch-failure handling.
- Secured the pipeline end-to-end using S3 pre-signed URLs, protecting data in transit and at rest.

Real-Time Streaming Inference Pipeline — Kafka, EventBridge, Amazon SageMaker, PySpark

Jun – Aug 2025

- Architected a real-time streaming inference pipeline that ingested Kafka-streamed data and served predictions through a production Amazon SageMaker endpoint, achieving p99 latency under 100ms.
- Configured SageMaker endpoint auto-scaling policies triggered on Invocations count, absorbing intermittent traffic bursts while sustaining consistent low-latency performance.
- Orchestrated Amazon EventBridge to trigger Amazon SageMaker Processing Jobs on Spark-based custom container images for large-scale data pre-processing, cutting operational costs and processing time 60-70% versus an AWS Lambda-based approach.

Solar Panel Maintenance Agent — Amazon Bedrock, OpenSearch Serverless, Lambda, DynamoDB

Mar – Apr 2025

- Built an agentic assistant on Amazon Bedrock with long-term session memory, summarizing each session and feeding it back into agent memory to maintain context across interactions, used internally for demo and training purposes.
- Equipped the agent with two tool-calling capabilities — an OpenSearch Serverless retrieval tool over internal S3 documents for solar panel installation/maintenance guidance, and a Lambda-backed API tool (/get_ticket_status, /open_ticket) connected to DynamoDB customer support data.
- Reduced customer support resolution time 40-50% versus manual, brute-force troubleshooting tools.

Instruction Fine-Tuning for Financial Document Understanding — SageMaker JumpStart, FLAN-T5, BERT

Jan – Mar 2025

- Fine-tuned FLAN-T5-XL and BERT pre-trained models via Amazon SageMaker JumpStart using instruction fine-tuning on proprietary financial documents, delivering a site-level, customer-specific model for finance-domain document understanding.
- Curated instruction-style prompt-response training pairs and tuned hyperparameters to align model outputs with domain-specific financial terminology and document formats.

RAG-Based Contextual Chatbot — Amazon Bedrock & OpenSearch

Oct – Dec 2024

- Boosted query resolution rates 80% by building a Retrieval-Augmented Generation chatbot on Amazon Bedrock Knowledge Bases powered by Claude models.
- Eliminated 100% of manual ingestion effort by orchestrating automated S3-to-Bedrock pipelines for 1,000+ product documents.
- Cut manual document look-up time 70% through semantic indexing (cosine-similarity embeddings) in Amazon OpenSearch Serverless.
- Elevated response relevance and production-readiness using metadata filtering, chunking optimization, re-ranking, and LLM evaluation metrics (context relevance, faithfulness).

PUBLICATIONS

- Language Variety Identification with True Labels — LREC-COLING 2024
- Privacy Preservation and Signature Aggregation for Medical Data — ICRTAC 2019